	7.2. PRODUCTION PLANNING	
	Document no: ASMF7.2.1	
Compiled by:	Sifiso Nyambi - Laboratory Manager	
Approved by:	Nonhlanhla Msibi - Operations Manager	

Production Plan

During the production planning of Proficiency Testing Scheme, homogeneity test is conducted (10 random samples are extracted from the sample/material product, tested for Quick Ash%). The samples are retained for determination of stability test for a period of 4 (four) weeks. The material retained after the PT production is packaged for Certified reference Material production. The goal of the production plan is to design the workflow that minimises wastage, attains PT and RM samples that have achieved sufficient homogeneity and stability requirements.

1. General Information

One of the Certified Reference Material to be produced is defined in table below:

Field	Entry
CRM Name	GA CRM AS002
Unique batch Number	CRM AS002 2025-05
Intended Use	Calibration, Control Sample, and Training of staff
Matrix type	Pulverised Coal
Properties to be assigned	ISO Ash, ISO Volatile Matter, Calorific Value, and Total Sulphur
Production Start date	2025-04-29
Production End Date	-
Responsible Person	Sifiso Nyambi
Approved by	Nonhlanhla Msibi

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2. Materials and Equipment

A list of materials and equipment to be used in the production of the GA CRM AS002 are tabled and listed as follows:

NB: The supplier of the materials will vary due to the required CRM quality to be produced. However, it is required that a record of list of suppliers of coal associated which a specific CRM be maintained at the laboratory. This is to ensure that the same CRM will be produced that is consistent with the initial quality produced.

NB: All other equipment used for the preparation and testing, the validation or equipment maintenance are specified in the respective coal testing methods ie. Procedure on sample preparation of Hard coal has detailed how to use crushers and efficiency requirements needed to produce coal size material required.

Item	Description/ Specs	Supplier	UoM	Serial Number	Qualification Service
Raw Material	Bituminous coal sample	Elandsfontein Mine	-	-	Approved
Crushers	25,10,3 (mm)	MegaSize	-	202203331 SC, AS8809, A124119; respectively	Cleaned and verified
Pulveriser	212 µm	MegaSize	-	22L90713190645	Cleaned and verified
Platform Scale	0.01 Kg	Darvan Brenco	±0.03 kg	120330150	Calibrated
Top Pan Balance	0.1 g accuracy	AE Adam	±0.3 g	AEA90242156	Calibrated
IM Moisture Oven 50L	Maintain a temperature of 107°C	Darvan Brenco	±1.0 °C	DOV53A/210325/40	Calibrated
Mass pieces	1 to 100 (g)	Micron	±0.00014	OB WS1.1R	Calibrated
Ash Furnace	Maintain 815±10°C for at least 1 hour	Micron (ATS)	±4 °C	ATS/210325/7A	Calibrated
ISO Volatile	900°C±5°C	Micron (ATS)	±3 °C	ATS/210325/59V	Calibrated

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Item	Description/ Specs	Supplier	UoM	Serial Number	Qualification Service
Matter Furnace					
Packaging Material	±150 g	Foamy	-	-	Cleaned
Homogeneity Drum	-	-	-	-	Cleaned

3. Processing Plan

Step	Description	Equipment used	Environmental Conditions	Acceptance criteria
1	Air drying	-	Atmospheric temperature	Sample should be visibly dry
2	Stage Crushing	25,10,3 mm Crushers	Room temperature	Crusher Efficiency ≥90%
3	Pulverise	Serial no: 22L90713190645	Room temperature	Pulveriser efficiency ≥95%
4	Homogeneity Drum	-	-	¬ 4 hours
5	Homogeneity Validation (analysis)	ISO Ash Furnace	19-26 °C	ANOVA, SSbetween<Check Value
6	Packaging	±150g	Room temperature	-
7	Storage	Storage room	≤ 20°C	Temperature <23°C

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4. Homogeneity and Stability Strategy

Assessment Type	Method	No of units	Parameter	Acceptance Criteria
Homogeneity	HOM-01	10	Quick Ash	ANOVA, SSbetween<Check Value
Short-term stability	STA-02	3 time points	Quick Ash	No trend or drift outside limits.
Long-term stability	STA-03	6 months, 12 months	Calorific Value and Q-ash	Deviation should not be >0.3MJ/kg and 3% of assigned value.

NB: A monthly proficiency testing report has recorded the respective homogeneity and statibility test of respective CRM. A long term stability test record is maintained.

5. Measurement Procedures

Parameter	Method Reference	Equipment	Analyst(s)	Estimated UoM
Q-Ash	ISO 1171	Furnace	Keitumetse M	
Volatile Matter	ISO 562	Furnace	Lilo M	
Calorific Value	ISO 1928	Bomb Calorimeter	Nosipho H	
Inherent Moisture	SANS 5925	Drying Oven	Keitumetse M	

NB: The following authorised procedures are to be used:

- AS002IW Procedure on the conditioning of samples,

- AS003IW Procedure for the determination of Ash content

- AS006IW Procedure for the determination of Volatile Matter

- AS004IW Procedure on the determination of Inherent Moisture

Effective date	Rev status	Next review date
07/2025	00	07/2028

- AS007IW Procedure on the Calorific Value determination.

6. Risk Assessment

Risk Description	Impact	Mitigation strategy
Power outage during analysis	Low	Backup generator
Balance drift	Medium	Verification with calibrated mass pieces at the beginning of every shift.
Sample Swap	High	Prepare one sample at a time, only after packaging has been completed then the next sample can be prepared.

7. Records and Approval

Record Type	File Name/ Location	Responsible Person	Retention Period
Batch Processing	Form	Laboratory Supervisor	5 years
Analytical Results Sheet	LIMS	Laboratory Supervisor	5 years
Homogeneity Test report and Stability Test report	[can be retrieved from within the final PT report] or website	Laboratory Manager/Supervisor	10 years
Calibration Certificates	Technical File	Lab Manager	5 Years